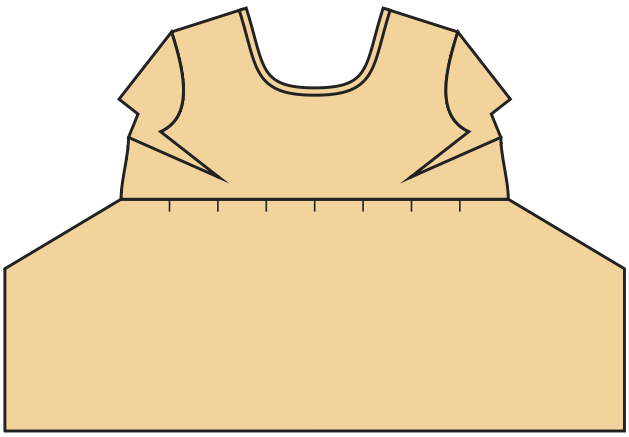
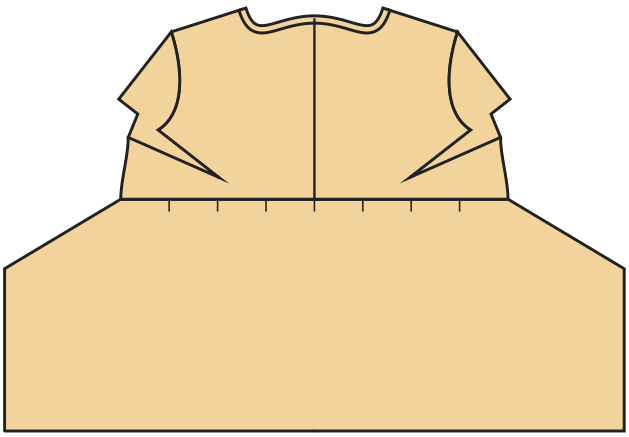
Dress with gathered skirt

doll 45 cm (made to measure) · wide fit, Woven (0 % stretch)

6 parts · 24 steps · advanced · about 3.5 h



Front view



Back view

Sewn with: Sewing machine



Sewing machine



Iron

Read the steps once before you start. Every picture is drawn from your pattern pieces. Sew a test garment from cheap fabric first and adjust before you cut the good fabric.

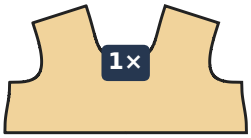
Parts



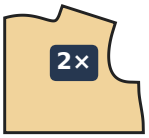
E · Skirt front (cut on fold)



F · Skirt back



A · Front (cut on fold)



B · Back



C · Sleeve



D · Neck binding (bias)

Seam allowance: included: 0.6 cm · hem 0.6 cm, 1.0 cm

Material

Main fabric
0.20 m at 150 cm width

Velcro
The back opens completely: close it with narrow Velcro (1 cm) or 2–3 small snaps.

Sewing machine
straight stitch 2.5 mm; wide zigzag 4 × 1.5 mm over the edge; straight stitch 3 mm; 2 rows of the longest straight stitch, loose ends

Iron
steam, cotton setting

Polyester sewing thread
In the colour of the fabric.

Pins or clips
Pin across the seam line.

Chalk or a fabric pen
For notches, labels and placement lines.

Colours in this booklet

Main fabric
right side · wrong side

Sew now (machine)

Sew by hand

Already sewn

Fold / press line

Marking (chalk)

Pin

Notch: match them

Right/wrong side of the fabric = the side of the fabric. Left/right (as worn) = seen from the child wearing it: the left front leg is on the right when you look at the child.

Basics: Sewing machine

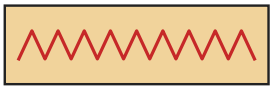
Steps are written for a sewing machine with zigzag. Have an overlocker or a coverstitch machine? Tell the generator which machines you own and the steps use them.

Basics: the sewing machine



Straight stitch

Seams in woven fabric, 2.5 mm; backstitch at the start and the end.



Zigzag

Knit seams (narrow) and finishing the edges (wide).

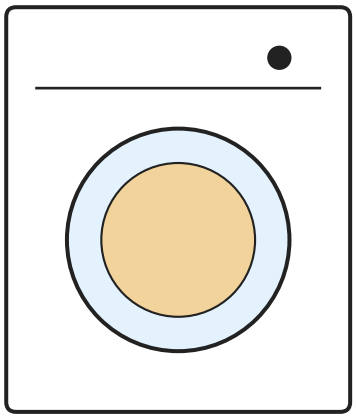


Topstitching

Right side up, with the same distance to the edge all the way; use the edge of the presser foot as a guide.

1 Pre-wash the fabric

1. Wash and dry it the way you will wash the garment later, so it shrinks now, not after sewing.



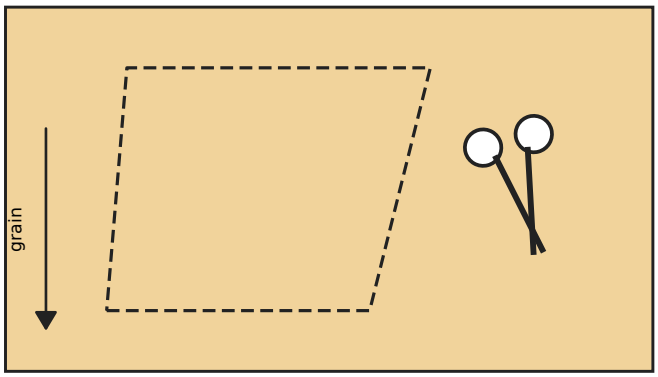
30–40°

2 Cut out all pieces

Scissors

1. Follow the cutting layout
2. Grainline arrows parallel to the selvage.

Tip: Cut both layers at once with long strokes; keep the fabric flat on the table.

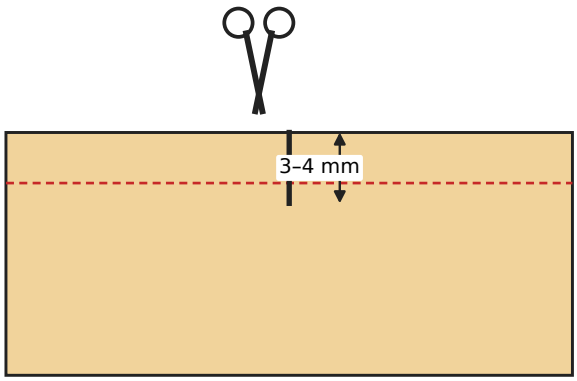


3 Snip the notches

Chalk

1. Cut 3–4 mm into the seam allowance at every notch.

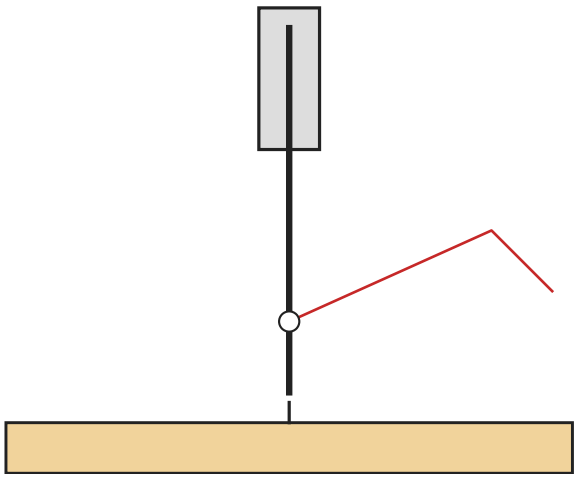
⚠ Important: Never deeper than half the seam allowance.



4 Set up the machine

1. Universal needle 80, polyester thread.
2. Sew a test seam on a scrap first.

Tip: See the basics page at the front of the booklet.



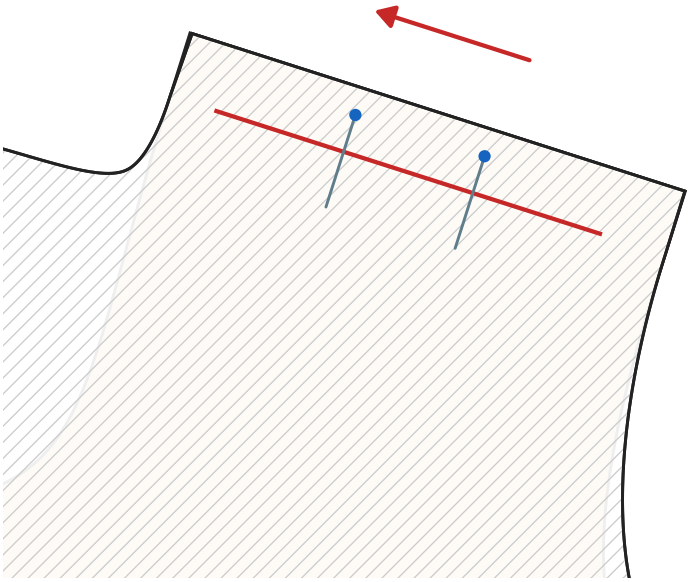
5 Sew the shoulder seams

Sewing machine

Pins

1. Lay front and back right sides together, shoulder edges level, and pin.
2. Sew with 6 mm seam allowance from the neckline to the armhole.
3. Press the allowances towards the back.

Sewing machine: straight stitch 2.5 mm



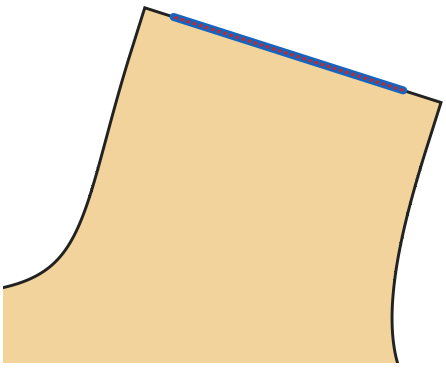
2×

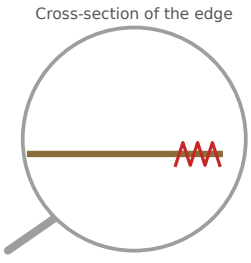
6 Finish the shoulder seams

Sewing machine

1. Finish both allowances together, from the right side of the upper layer.
2. Press them towards the back.

Sewing machine: wide zigzag 4 × 1.5 mm over the edge





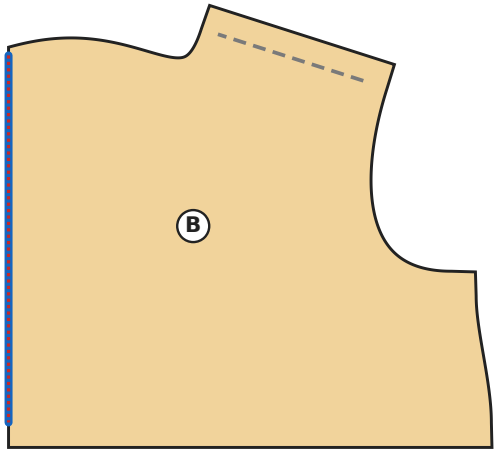
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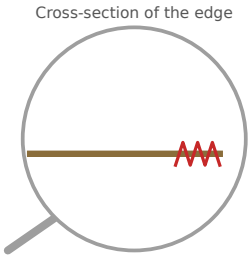
7 Finish the opening edges

Sewing machine

1. Finish both centre back edges separately.

Sewing machine: wide zigzag 4 × 1.5 mm over the edge





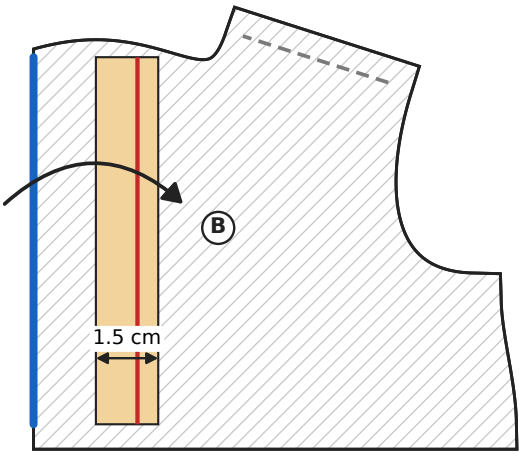
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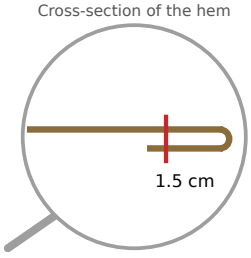
8 Turn in the opening edges

Sewing machine

1. Fold the opening edge 1.5 cm to the wrong side and press.
2. Stitch it down close to the finished edge.

Sewing machine: straight stitch 3 mm

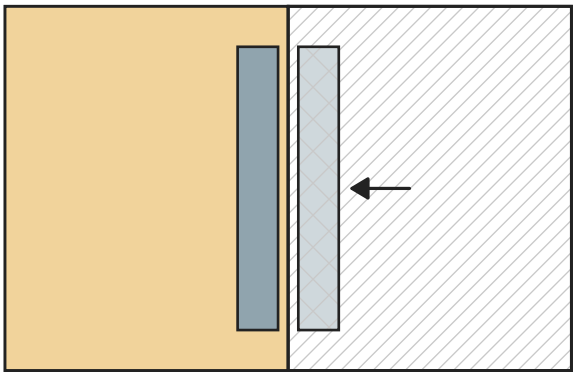




2×

9 Sew on the Velcro

Sewing machine



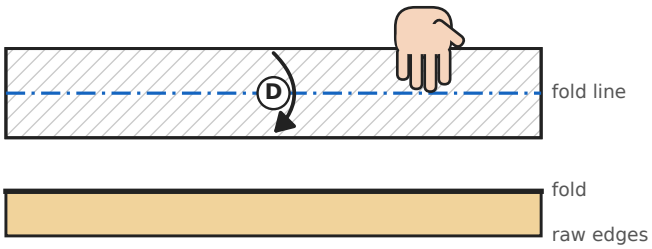
1. Lay the right back over the left back so the edges overlap at the centre back.
2. Sew the soft side to the right side of the left back and the hook side to the inside of the right back, stitching round each strip (or sew on 2–3 snaps).

Sewing machine: straight stitch 3 mm

⚠ Important: The garment opens completely: easy dressing.

10 Fold the binding in half

Iron

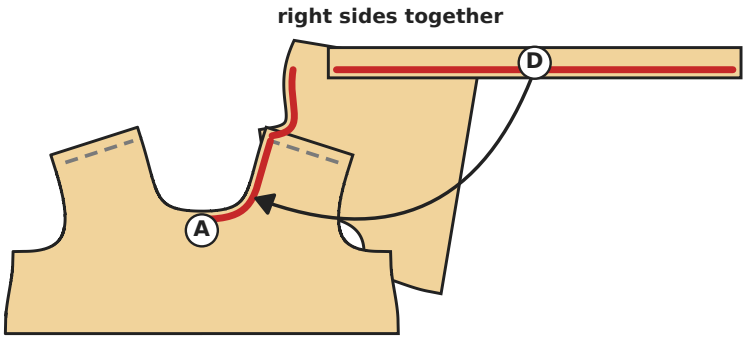


1. Fold the bias strip lengthwise wrong sides together, raw edges level, and press.
2. Shape it into a curve with the iron.

Iron: steam, cotton setting

11 Sew the neck binding on

Sewing machine · Pins

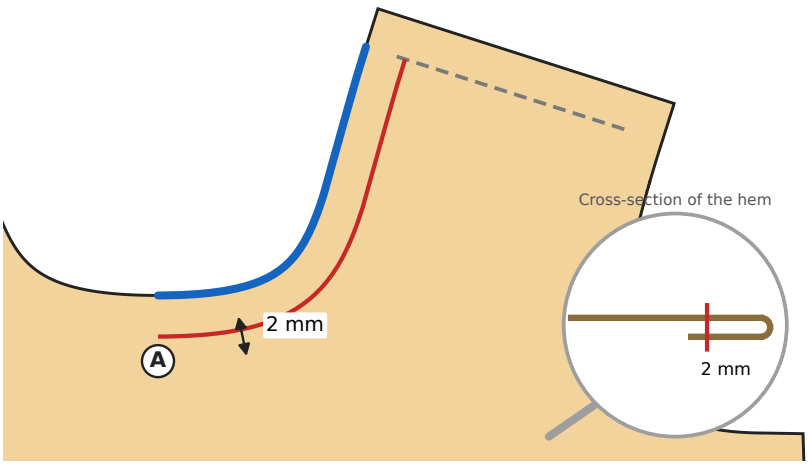


1. Lay the binding on the right side of the neckline, raw edges level
2. Let each end stick out past the opening edge by the allowance.
3. Pin without stretching.
4. Sew with 6 mm seam allowance.
5. Trim the allowance to half and clip it in the curves.

Sewing machine: straight stitch 2.5 mm

12 Stitch the binding down

Sewing machine



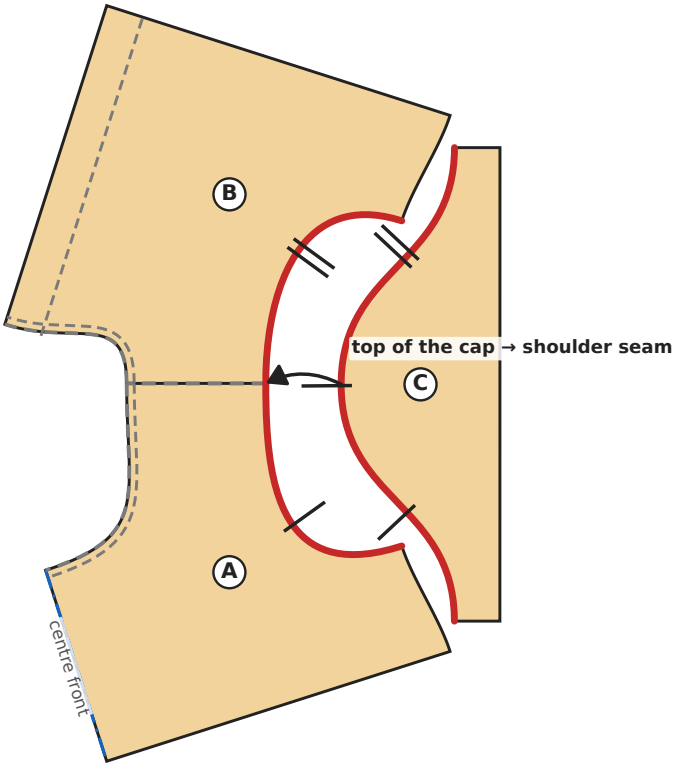
1. Fold the ends in.
2. Turn the binding to the inside over the allowance and press.
3. Stitch it down close to the fold.

Sewing machine: straight stitch 3 mm

Tip: Press first: the binding then rolls inside by itself.

13 Set in the sleeves

Sewing machine · Pins



1. Open the body out flat, right side up.
2. Lay the sleeve right side down on the armhole: top of the cap on the shoulder seam, single notch on the front, double notch on the back.
3. Pin from the middle to both ends, easing in the extra length of the cap.
4. Sew with 6 mm seam allowance, sleeve underneath
5. Press the allowances towards the sleeve.

Sewing machine: straight stitch 2.5 mm

⚠ Important: Sleeve on the bottom: the feed dogs ease the cap in without pleats.

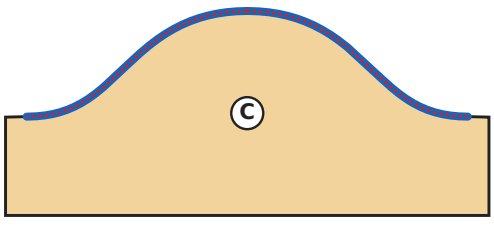
2×

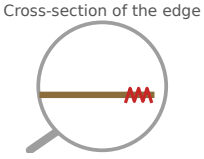
14 Finish the armhole seams

Sewing machine

1. Finish both allowances together, from the right side of the upper layer.
2. Press them towards the back.

Sewing machine: wide zigzag 4 × 1.5 mm over the edge

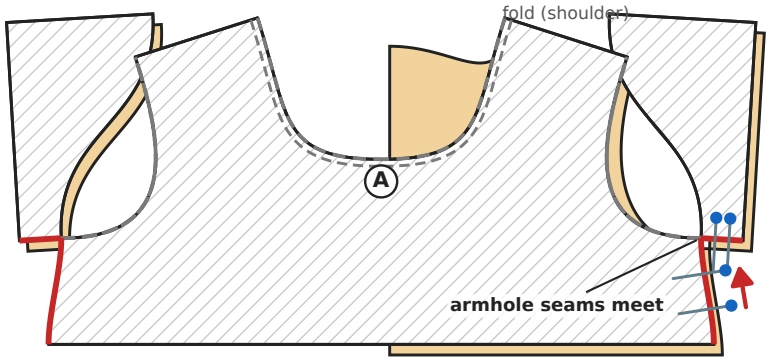




2×

15 Sew side and sleeve seams

Sewing machine · Pins



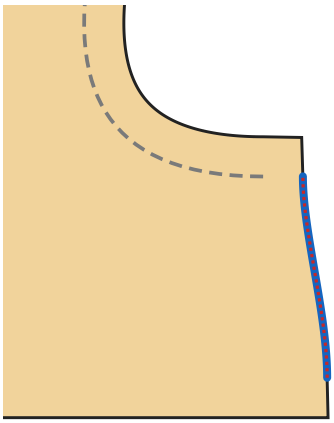
1. Fold the garment right sides together at the shoulders, the sleeves folded in half lengthwise.
2. Pin side and sleeve seam so that the armhole seams meet exactly at the underarm.
3. Sew in one go with 6 mm seam allowance, from the hem over the underarm to the sleeve hem.
4. Press the allowances towards the back.

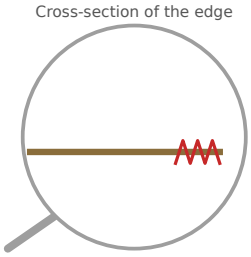
Sewing machine: straight stitch 2.5 mm

2×

16 Finish the side seams

Sewing machine





1. Finish both allowances together, from the right side of the upper layer.
2. Press them towards the back.

Sewing machine: wide zigzag 4 × 1.5 mm over the edge

2×

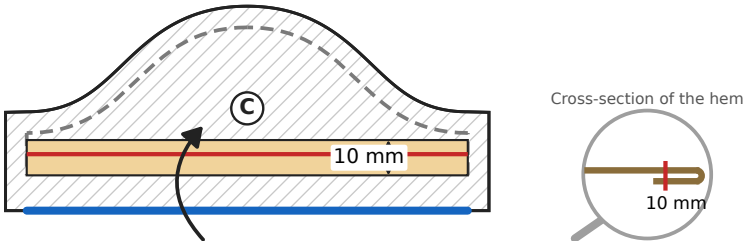
17 Hem the sleeves

Sewing machine

1. Fold the sleeve hem 10 mm to the wrong side and press.
2. Sew it down 6 mm from the fold, from the right side.

Sewing machine: straight stitch 3 mm

Tip: Slide the sleeve over the free arm of the machine.



2×

18 Sew the skirt centre back seam

Sewing machine

Pins

1. Lay the skirt back pieces right sides together, centre back edges level.
2. Sew with 6 mm seam allowance from the hem up to the notch.
3. Above the notch the seam stays open and continues the back opening.

Sewing machine: straight stitch 2.5 mm

⚠ Important: Backstitch firmly at the notch: the opening pulls there.



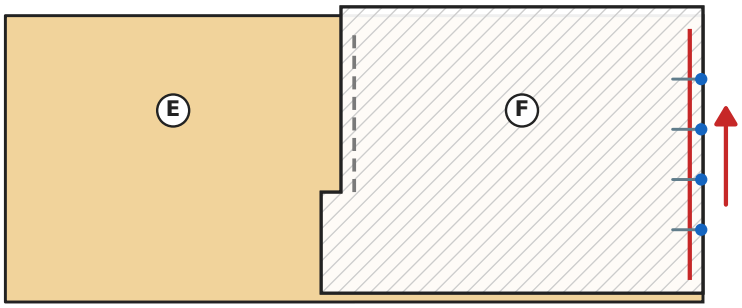
19 Sew the skirt side seams

Sewing machine

Pins

1. Lay skirt front and back right sides together, side edges level.
2. Sew with 6 mm seam allowance from the waist to the hem.
3. Press the allowances towards the back.

Sewing machine: straight stitch 2.5 mm

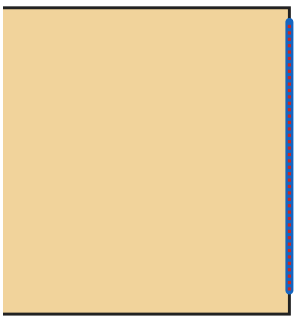


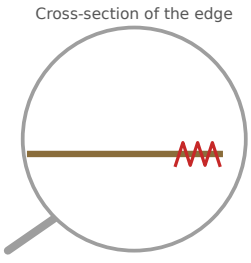
2×

20 Finish the skirt side seams

Sewing machine

Sewing machine: wide zigzag 4 × 1.5 mm over the edge

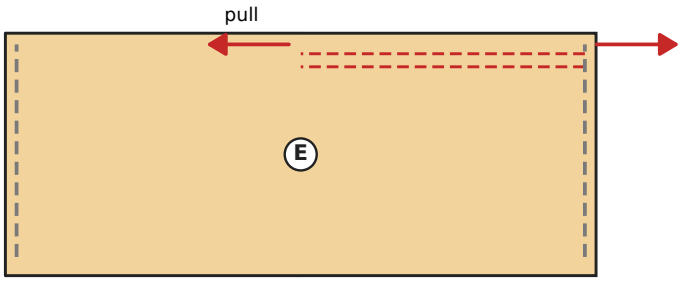




2×

21 Gather the skirt top edge

Sewing machine



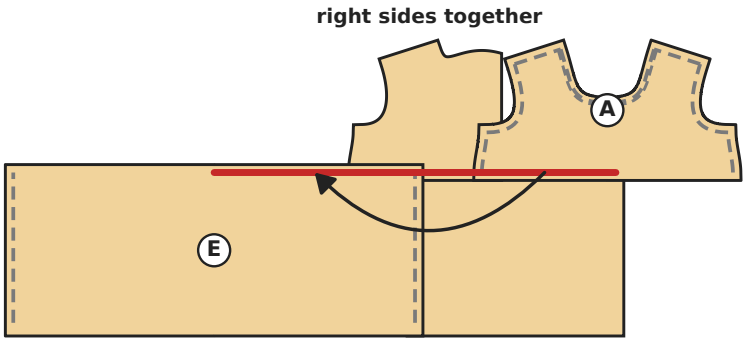
1. Sew two rows of the longest stitch inside the seam allowance, 5 mm apart
2. Leave long thread ends.
3. Pull the bobbin threads until the edge matches the bodice waist (gather ratio 1.6).
4. Spread the gathers evenly and knot the threads.

Sewing machine: 2 rows of the longest straight stitch, loose ends

Tip: Pull from both ends towards the middle: the threads break less easily.

22 Sew skirt and bodice together

Sewing machine · Pins



1. Turn the skirt wrong side out and slip the bodice into it, right side out, so the right sides lie together and the waist edges are level.
2. Match the side seams and the centre front and back, and pin.
3. Sew with 6 mm seam allowance, skirt on top.
4. Press the allowances towards the bodice.

Sewing machine: straight stitch 2.5 mm

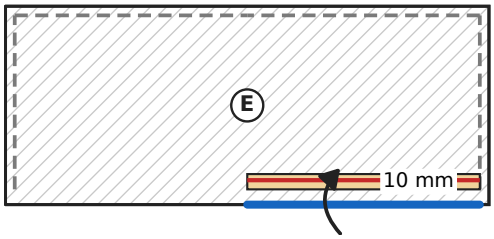
Tip: Sew over the gathers slowly and keep them straight with a pin tip.

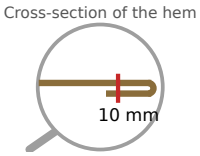
23 Sew the hem

Sewing machine

1. Fold the hem 5 mm to the wrong side and press, then fold it again by the same amount.
2. Sew it down close to the inner fold.

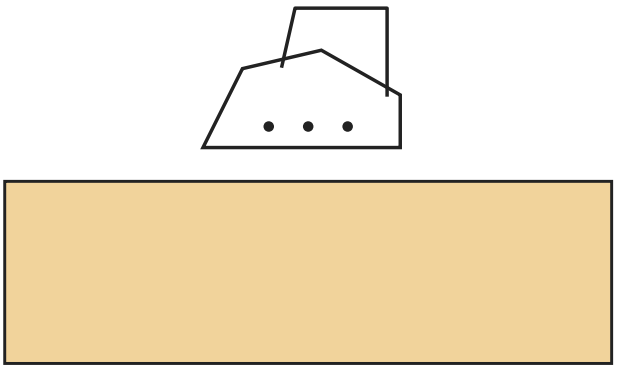
Sewing machine: straight stitch 3 mm





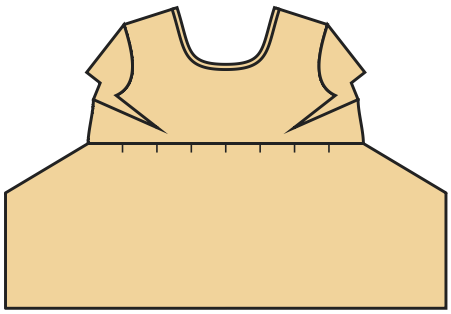
24 Final pressing

1. Press all seams.

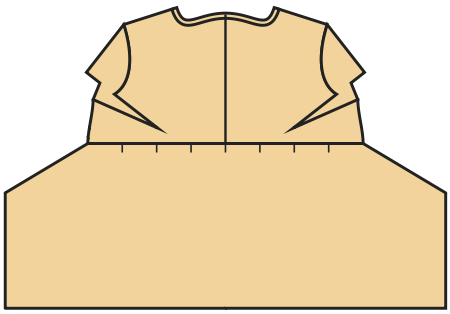


Done!

After the test garment: Try the test garment on the doll: does it go over the head, does the opening close, is the length right? Adjust the measurements and print the pattern again before you cut the good fabric.



Front view



Back view

Dress with gathered skirt · doll 45 cm (made to measure)

wide fit, Woven (0 % stretch) · original parametric draft. The pattern is yours to use.

> **Important:** With a new pattern, first sew a **test garment** from inexpensive fabric with similar stretch and check the fit.

Files

- PDF A4 tiles to glue (3 pages + overview): `doll_dress_45cm_a4_tiled.pdf`
- Preview A4 tiles: `doll_dress_45cm_a4_preview.png`
- SVG master 1:1: `doll_dress_45cm_master_1to1.svg`
- Cutting layout: `cutting_layout.png`
- Sewing instructions (PDF): `sewing_instructions.pdf`
- Sewing instructions (Markdown): `sewing_instructions.md`

Printing

- Always print at **100 % / "Actual size"** (not "Fit to page").
- Measure the **5 × 5 cm** control square before you cut.
- Plotter (610 mm roll): the page is 600 mm wide in portrait. In the print dialog choose a custom paper size with exactly this width and length, and turn rotation/scaling off.

Body measurements

Body measurement	Value	Source
Body height	45.0 cm	user
Chest	30.0 cm	user
Waist	28.0 cm	user
Hip	31.0 cm	user
Head	32.0 cm	user
Neck	15.1 cm	derived
Inseam	14.0 cm	user
Side length	22.0 cm	user
Arm length	12.0 cm	user
Back length	9.0 cm	derived
Shoulder width	3.2 cm	derived
Thigh	18.0 cm	derived
Ankle	11.4 cm	derived

Finished measurements (virtually sewn)

Finished measurement	Value
Chest width	36.9 cm
Hem width	60.9 cm
Dress length from neck point	20.7 cm
Sleeve length	4.2 cm
Sleeve opening	12.5 cm
Upper-arm width	12.5 cm
Neckline circumference	20.4 cm
Waist seam (bodice)	38.1 cm
Skirt top edge (ungathered)	60.9 cm
Skirt length	11.4 cm

Pattern pieces

Piece	Cutting	Fabric	Seam allowance
Front	cut 1 on fold	Main fabric	seam allowance 0.6 cm included; Neckline: 0.4 cm
Back	cut 2 (mirrored pair)	Main fabric	seam allowance 0.6 cm included; Back opening (turn under): 1.5 cm; Neckline: 0.4 cm
Sleeve	cut 2 (mirrored pair)	Main fabric	seam allowance 0.6 cm included; Sleeve hem: 1.0 cm
Neck binding (bias)	cut 1	Main fabric	seam allowance 0.4 cm included
Skirt front	cut 1 on fold	Main fabric	seam allowance 0.6 cm included; Hem: 1.0 cm
Skirt back	cut 2 (mirrored pair)	Main fabric	seam allowance 0.6 cm included; Hem: 1.0 cm; Back opening (turn under): 1.5 cm

Fabric needed

- **Main fabric:** about 0.20 m at 150 cm width (single layer, incl. 10 % for shrinkage; folded would need 0.30 m)
- Doll clothes: narrow seam allowances (6 mm). Sew with a short stitch (1.5–2 mm) and finish woven edges with pinking shears, a narrow zigzag or a little fray stop.
- Thin, soft fabrics work best (cotton poplin, batiste, old T-shirts); thick fabric makes the tiny seams bulky.
- The back opens completely: close it with narrow Velcro (1 cm) or 2–3 small snaps.

Cutting layout: see `cutting_layout.png` (grainline arrow parallel to the selvage, "flipped" = place the piece face down, "×2" = cut through both layers of the folded fabric, which gives a mirrored pair, "(unfolded)" = cut-on-fold piece shown opened out).

Cut on fold (Front, Skirt front): the paper piece is only one half. Its fold edge (dash-dot line, bracket arrows "PLACE ON FOLD") is the centre of the piece: it gets no seam allowance and is never cut. Where the cutting layout shows a piece "(unfolded)" on a single layer, fold the fabric along the grain just wide enough for the piece and place the fold edge on that fold, or trace the paper piece, flip it over along the fold edge and trace the second half.

Validation report

Sewability check

Area	Check	Target	Actual	Δ	Result	Note
Seam pairs	Side seam front ↔ back	3.1 cm	3.1 cm	+0.0 mm	□ OK	
Seam pairs	Armhole ↔ sleeve cap (ease)	ease 3–8 mm	13.8 cm ↔ 14.3 cm	+5.4 mm	□ OK	Woven: 3–8 mm
Seam pairs	Shoulder seam front ↔ back	3.7 cm	3.7 cm	+0.0 mm	□ OK	
Seam pairs	Sleeve seam (front ↔ back)	1.2 cm	1.2 cm	+0.0 mm	□ OK	
Seam pairs	Neck binding ↔ neckline	20.4 cm	20.4 cm	+0.0 mm	□ OK	
Seam pairs	Waist seam bodice ↔ skirt (gathered)	40–100 % of 60.9 cm	38.1 cm = 63 %	-228.4 mm	□ OK	gather ratio 1.6 (allowed 1–2.5)
Seam pairs	Skirt side seam front ↔ back	11.4 cm	11.4 cm	+0.0 mm	□ OK	
Seam pairs	Skirt centre back seam (left ↔ right)	7.4 cm	7.4 cm	+0.0 mm	□ OK	open for the top 4.0 cm (notch)
Angles/transitions	Underarm: side seam ↔ armhole front	90°	91.9°	+1.9°	□ OK	
Angles/transitions	Underarm: side seam ↔ armhole back	90°	92.2°	+2.2°	□ OK	

Area	Check	Target	Actual	Δ	Result	Note
Angles/transitions	Shoulder ↔ armhole (front + back)	180°	180.6°	+0.6°	✅ OK	
Angles/transitions	Sleeve cap top (smooth)	180°	178.2°	-1.8°	✅ OK	
Angles/transitions	Sleeve underarm ↔ cap front	90°	90.9°	+0.9°	✅ OK	
Angles/transitions	Shoulder ↔ neckline (front + back)	180°	180.9°	+0.9°	✅ OK	
Angles/transitions	Neckline at centre front	90°	90.4°	+0.4°	✅ OK	
Angles/transitions	Neckline at centre back	90°	89.0°	-1.0°	✅ OK	
Angles/transitions	Hem at centre front	90°	90.0°	+0.0°	✅ OK	
Angles/transitions	Hem ↔ side seam front	90°	88.8°	-1.2°	✅ OK	
Angles/transitions	Skirt hem ↔ side seam	90°	90.0°	+0.0°	✅ OK	
Angles/transitions	Skirt side seam ↔ waist	90°	90.0°	+0.0°	✅ OK	
Angles/transitions	Skirt waist ↔ centre	90°	90.0°	+0.0°	✅ OK	
Measurements	Chest circumference pattern = body + ease	36.6 cm	36.9 cm	+0.8 %	✅ OK	body 30 cm + +22 % ease
Measurements	Head fits through the back opening	≥ 33.6 cm	38.1 cm	+45.0 mm	✅ OK	neckline + both edges of the back opening ≥ head circumference + 5 %
Measurements	Dress goes on through the back opening	≥ 32.6 cm	46.0 cm	+134.9 mm	✅ OK	waist seam + both edges of the skirt opening ≥ hip + 5 %
Measurements	Dress length plausible	14.9 cm – 33.5 cm	20.7 cm	+0.0 mm	✅ OK	from the neck point
Geometry	Front: outline & seam allowance	closed, SA present	ok		✅ OK	seam allowance 0.6 cm included; Neckline: 0.4 cm
Labels	Front: grainline & cutting	present	cut 1 on fold		✅ OK	Main fabric
Geometry	Back: outline & seam allowance	closed, SA present	ok		✅ OK	seam allowance 0.6 cm included; Back opening (turn under): 1.5 cm; Neckline: 0.4 cm
Labels	Back: grainline & cutting	present	cut 2 (mirrored pair)		✅ OK	Main fabric
Geometry	Sleeve: outline & seam allowance	closed, SA present	ok		✅ OK	seam allowance 0.6 cm included; Sleeve hem: 1.0 cm
Labels	Sleeve: grainline & cutting	present	cut 2 (mirrored pair)		✅ OK	Main fabric
Geometry	Neck binding (bias): outline & seam allowance	closed, SA present	ok		✅ OK	seam allowance 0.4 cm included
Labels	Neck binding (bias): grainline & cutting	present	cut 1		✅ OK	Main fabric
Geometry	Skirt front: outline & seam allowance	closed, SA present	ok		✅ OK	seam allowance 0.6 cm included; Hem: 1.0 cm

Area	Check	Target	Actual	Δ	Result	Note
Labels	Skirt front: grainline & cutting	present	cut 1 on fold		□ OK	Main fabric
Geometry	Skirt back: outline & seam allowance	closed, SA present	ok		□ OK	seam allowance 0.6 cm included; Hem: 1.0 cm; Back opening (turn under): 1.5 cm
Labels	Skirt back: grainline & cutting	present	cut 2 (mirrored pair)		□ OK	Main fabric
Geometry	Back: turned allowance lies flat	≤ 0.5 mm	0.0 mm		□ OK	Back opening (turn under): corners mirrored at the fold line
Geometry	Sleeve: turned allowance lies flat	≤ 0.5 mm	0.0 mm		□ OK	Sleeve hem: corners mirrored at the fold line
Geometry	Front: cut-on-fold edge	straight, no SA, 90° corners, on grain	ok, corners ≤ 0.4° off 90°		□ OK	Centre (fold)
Geometry	Skirt front: cut-on-fold edge	straight, no SA, 90° corners, on grain	ok, corners ≤ 0.0° off 90°		□ OK	Centre (fold)
Labels	Front: notches	on the cutting line	1 notch mark(s)		□ OK	
Labels	Back: notches	on the cutting line	2 notch mark(s)		□ OK	
Labels	Sleeve: notches	on the cutting line	4 notch mark(s)		□ OK	1 at corners
Labels	Skirt back: notches	on the cutting line	2 notch mark(s)		□ OK	

Result: 45 OK, 0 warning(s), 0 error(s): the pattern is sewable.

Construction values

Override any of these with the option `construction`, values in mm (e.g. `{"construction": {"chest_width": 366}}`).

Name	Value	Derived from	Note
`chest_width`	36.6 cm	chest × (1 + 22 % ease)	
`neck_width_half`	3.5 cm	neck / 5 + 1.5 cm	
`front_neck_depth`	4.0 cm	neck_width_half + 1.5 cm	
`back_neck_depth`	0.5 cm	fixed	
`shoulder_length`	3.7 cm	shoulder + allowance for fit 'wide'	
`shoulder_slope`	17	fixed	
`armhole_depth`	6.3 cm	18 % of chest_width + 2.5 cm + 3 %, limited to option armhole_depth_max	
`body_length`	9.3 cm	option body_length	

Cutting layout · Dress with gathered skirt doll 45 cm (made to measure)

Main fabric: 150 cm wide, single layer
needed ≈ 0.20 m

